

LandCare Quick Handover

Purpose: explain how LandCare works today, how to maintain it with Codex, and how to replace Regrid with a URA-controlled contractor front end while keeping ArcGIS as the GIS platform.

1. System architecture and data flow

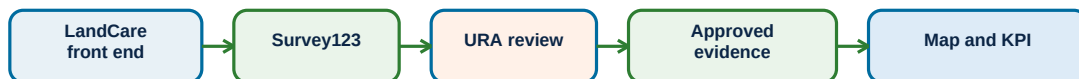
LandCare has three working layers:

- 1 **Data and ingestion.** Regrid survey exports and monthly assignment bundles are loaded by the GIS VM into PostgreSQL `gisdb` and published to ArcGIS Online.
- 2 **GIS services.** ArcGIS hosts the live survey, assignment, EPP parcel, council district, and approved evidence layers. These are the main browser data sources.
- 3 **Business application.** This repository publishes Map Monitor, KPI, contractor, and survey-submission pages through GitHub Pages. The browser joins assignments and survey evidence by normalized parcel number and period.

CURRENT REPORTING



URA-CONTROLLED INTAKE



Area	Current source or location	Business use
Survey evidence and comments	ArcGIS item 7a2e1d9bacba461296c54a63f104cf51, <code>regrid_surveys/FeatureServer/0</code>	Live photos, service dates, comments, field notes, and completion matching
Assignment denominator	Current item 0b4733cb5d204da6ab936c9f6d49e401 and history item df7d77eb57f14c68b717c2cf3cdaada4	Contractor, parcel, maintenance level, and reporting period
Current LandCare parcel universe	<code>gisdb_gis_epp_parcel_full</code>	Current URA-owned LandCare parcels and map geometry
Static fallback and finance	<code>docs/landcare/data/</code>	Fallback JSON and GeoJSON, finance totals, and refresh metadata
Public application	https://ura-gis.github.io/land-care-assurance/	Map Monitor, KPI, contractor portal, and submission page

The current completion numerator is the raw count of survey records whose normalized `parcelnumb` matches an assignment in the selected filter. It is not a unique-parcel count. Request Only records remain outside the Active completion denominator.

2. Replace Regrid with a URA-controlled front end

Use a staged replacement. Do not replace the reporting source and contractor workflow on the same day.

Stage 1: keep the custom front end and use Survey123 for submission

- Read active assignments from the public query-only ArcGIS current and history layers.
- Let the contractor select a period, organization, and parcel from the same map and list controls.
- Pass organization, parcel_number, address, and assignment_period into Survey123.
- Capture service date, work questions, comments, and photo attachments.
- Store new records as review_status = pending. Contractors cannot edit review fields.

Stage 2: validate and approve evidence

- Use a restricted Survey123 review view or Inbox for URA approval and rejection.
- Send new and edited records to the VM webhook.
- Validate assignment ID, parcel number, contractor, and period against the authoritative assignment layer.
- Mirror valid records into PostgreSQL and publish only approved parcel polygons and approved attachment links to a public read-only ArcGIS view.
- Never publish the raw Survey123 point layer, private review fields, tokens, or unapproved photos.

Stage 3: cut over from Regrid

- Run Regrid and the new path together for at least two service periods.
- Compare counts, parcel matches, contractors, dates, comments, photos, rejected records, and unmatched records.
- Keep the shared browser adapter contract: parcel number, period, contractor, service date, image URL, and canonical additional_notes.
- After business sign-off, point Map Monitor and KPI to the approved URA source, update metric documentation and tests, then retire the Regrid task and credentials.

If Survey123 later limits the contractor experience, replace only the form with a custom LandCare form. The browser should submit through a protected URA API. That service performs validation and writes to PostgreSQL or a secured ArcGIS feature service. Do not place publisher credentials in GitHub Pages or allow anonymous update and delete access to an authoritative layer.

3. Common Codex prompt

Work in the land-care-assurance repository. Read `HANDOVER.md`, `AGENTS.md`, `docs/landcare-architecture.md`, `docs/landcare-metrics-context.md`, and the relevant submission or VM runbook before editing.

Goal: [describe the business outcome and affected page].

Source: [ArcGIS item, service, PostgreSQL table, or published JSON].

Metric rule: completion uses raw assignment-matched survey records; Request Only is excluded from the Active denominator.

Before editing, state the source, field mapping, affected pages, tests, and rollback. Preserve live ArcGIS-first behavior, HTML escaping, URL validation, record ordering, evidence deduplication, and shared adapter cache versions. Do not add credentials or edit generated data by hand.

Implement the smallest complete change. Run Python tests, the survey-layer Node tests, Pages validation, and a read-only ArcGIS smoke test. Summarize the result, test evidence, and rollback path before commit and push.

4. Design system and release rules

Copy `docs/design-system/executive-bi.css` and the relevant markup from `example.html`. Wrap the new page in `.bi-dashboard`. Change branding through the `--bi-*` tokens instead of editing each component.

Use this page order: masthead, context and filters, decision heading, three to five KPIs, one main map or chart, and an operational ledger. Blue means primary analysis, navy means authoritative detail, orange means follow-up, and green means complete or stable. Always include a text label with color.

Every important number must show its source, period, denominator, and freshness. Keep controls keyboard accessible, normal text contrast at least 4.5:1, mobile stacking below 760 px, and clear loading, empty, stale, and error states.

Before release:

- Confirm ArcGIS layers are query-only where public and restricted where they contain review data.
- Run the repository tests and open Map Monitor, KPI, contractor, and survey-submission pages.
- Verify counts, comments, photos, field notes, filters, and the selected reporting period.
- Check the VM status JSON after scheduled workflow changes.
- Use a focused branch and pull request. Record source, denominator, affected page, test command, and rollback.

Canonical references: `HANDOVER.md`, `AGENTS.md`, `docs/landcare-architecture.md`, `docs/landcare-submission-and-evidence-flow.md`, `docs/survey123-landcare-network-setup.md`, `docs/landcare-metrics-context.md`, and `docs/design-system/README.md`.